

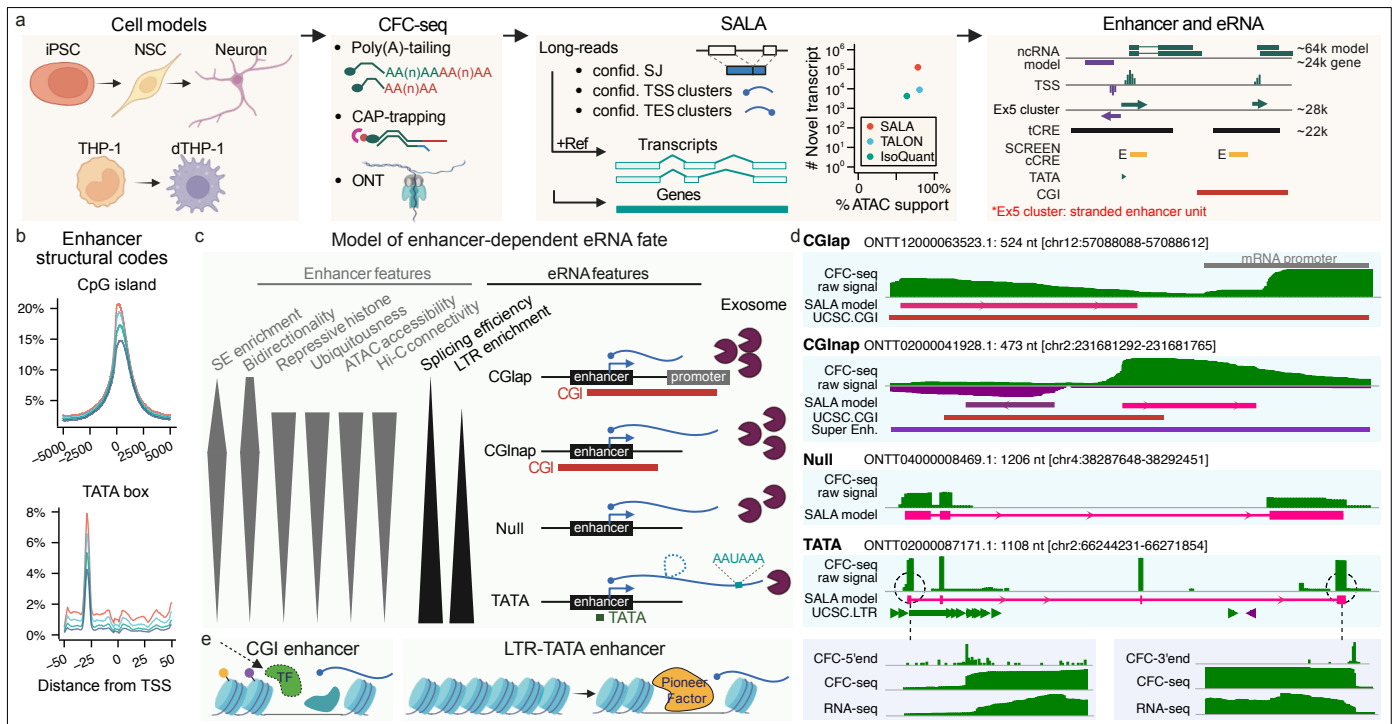


Figure 1 | CFC-seq and SALA assembly reveal a structural code for eRNA fate.

a, Overview of the CFC-seq, SALA assembly and enhancer annotation. Cap-trap full-length cDNA sequencing (CFC-seq) utilizes *in vitro* poly(A)-tailing and cap-trapping to capture diverse transcriptomes, including non-polyadenylated RNAs, across human differentiation models (Neurons and THP-1). The SALA assembler integrates CFC-seq long-reads with high-confidence splice junctions (SJ), transcription start site (TSS), and transcription end site (TES) clusters to generate precise transcript models. Comparison of SALA against standard assemblers (TALON, IsoQuant) reveals SALA's superior sensitivity in capturing biologically supported novel transcripts. Annotation of enhancers and eRNAs reveals ~24,000 eRNAs. Extended 5' end clusters (Ex5 clusters) represents stranded enhancer units in defining CpG island enhancer proximal to promoter (CGIap), CGI not proximal to promoter (CGIinap), TATA-box enhancer (TATA), and Null. **b**, Density plots showing the enrichment of CGIs and TATA-box motifs relative to the TSS of eRNAs. **c**, Schematic illustrating how enhancer structural features dictate RNA fate, including transcript length, splicing efficiency, polyadenylation and stability. **d**, Representative and simplified genomic tracks showing raw CFC-seq signal and SALA models across four eRNA classes. Transcript IDs refer to the eRNAs transcribed from plus strand. Bottom panels highlight the precise 5' and 3' end enrichment of CFC-seq compared to standard RNA-seq. **e**, Transcription activation from CGI enhancer and LTR-TATA enhancer. CGI enhancer is generally primed and ready to receive histone acetylation signal, while LTR-TATA enhancer is remodeled by NY-F and produce RNA without requiring histone acetylation.